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SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I T. Bhupal / 192425204 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T Bhupal

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	1.6	1.6	1.6
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	1.6	1.6	1.6
Application & Analysis	20%	1.6	1.6	1.6
Explanation & Clarity	15%	1.2	1.2	1.2

Total Marks 24 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

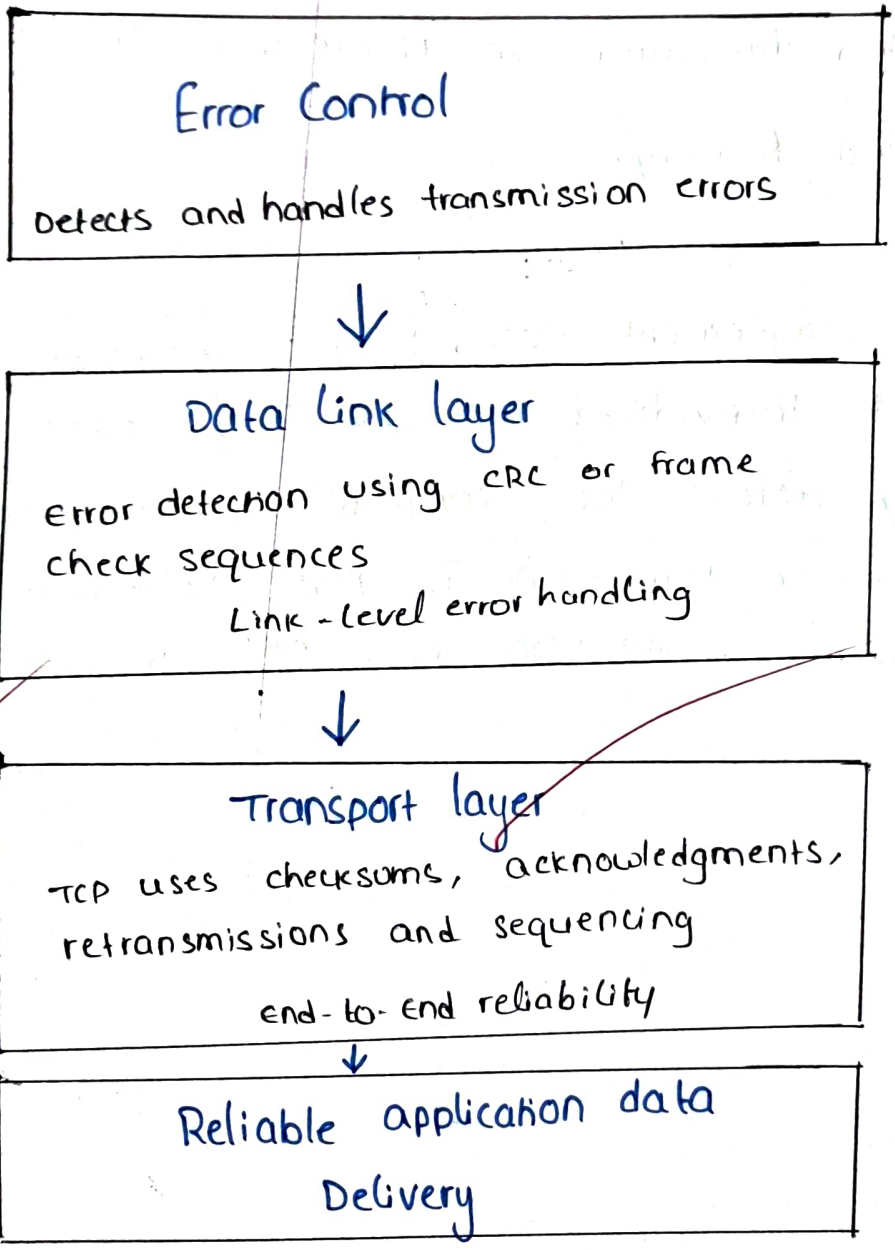
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application &	20%	Effectively applies	The application is	Basic	Unable to

Analysis		concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	correct but lacks depth in analysis	application with weak explanation	apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

SA0710
Computer Networks
01AT2

T. BHOPAL
192425243

Error Control in the Data Link and Transport Layers.

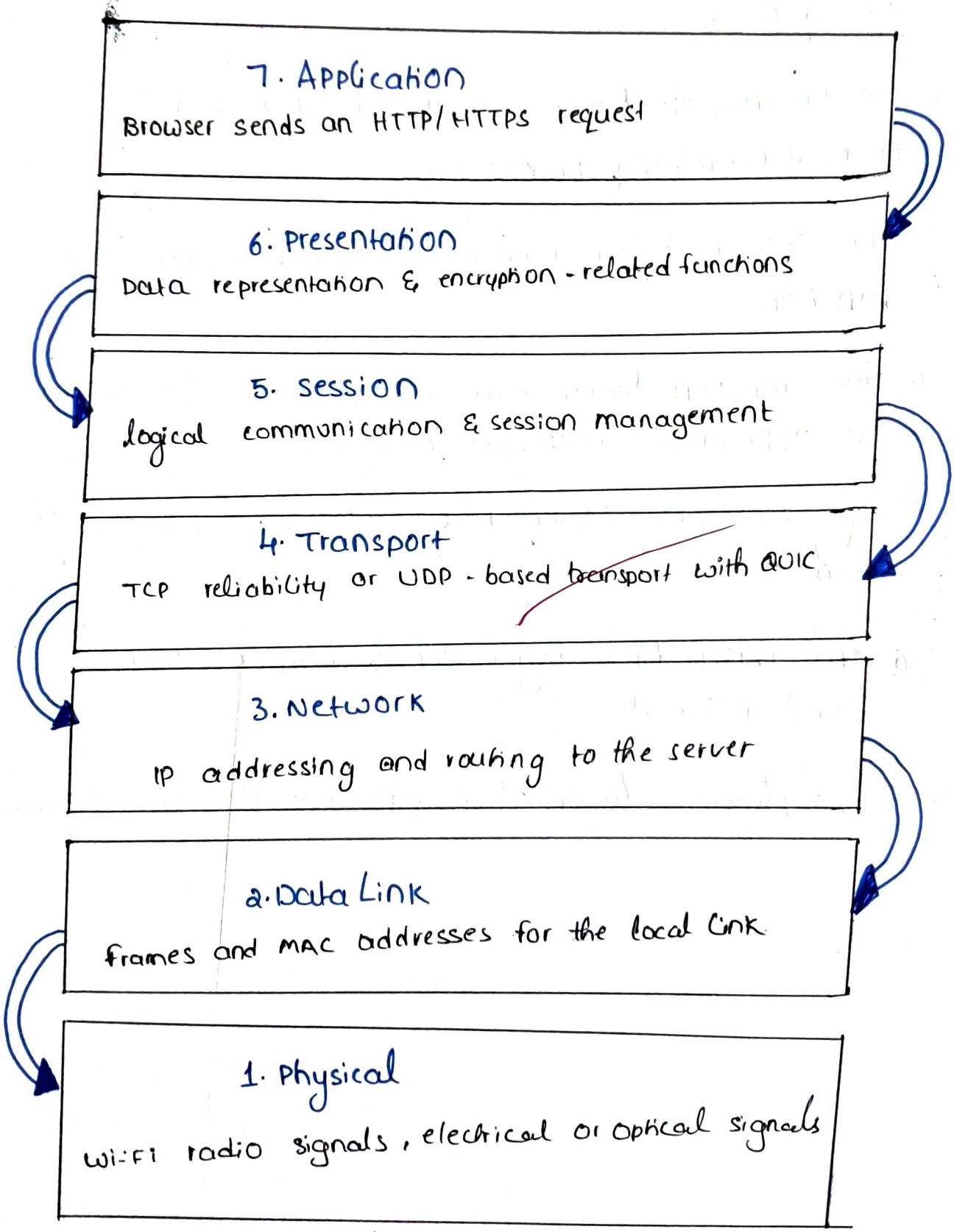


The Data Link layer detects corrupted frames on a link and may discard them or use link-level recovery mechanisms. The transport layer, particularly TCP, detects missing or corrupted data and can retransmits it to provide reliable end-to-end delivery.

These layers complement each other: link-level error handling addresses individual network links, while TCP handles reliability across the complete communication path. Note that error correction is not exclusively a Transport layer function, and UDP does not provide TCP-style

Web browsing using all OSI layers

Example : A user opens "https://www.google.com" in a web browser



web server receives request and sends a response

Explanation:

① The Application Layer provides web service through the browser

② the Presentation Layer concerns data formates & encryption-related processing.

③ the session layer represents session management functions

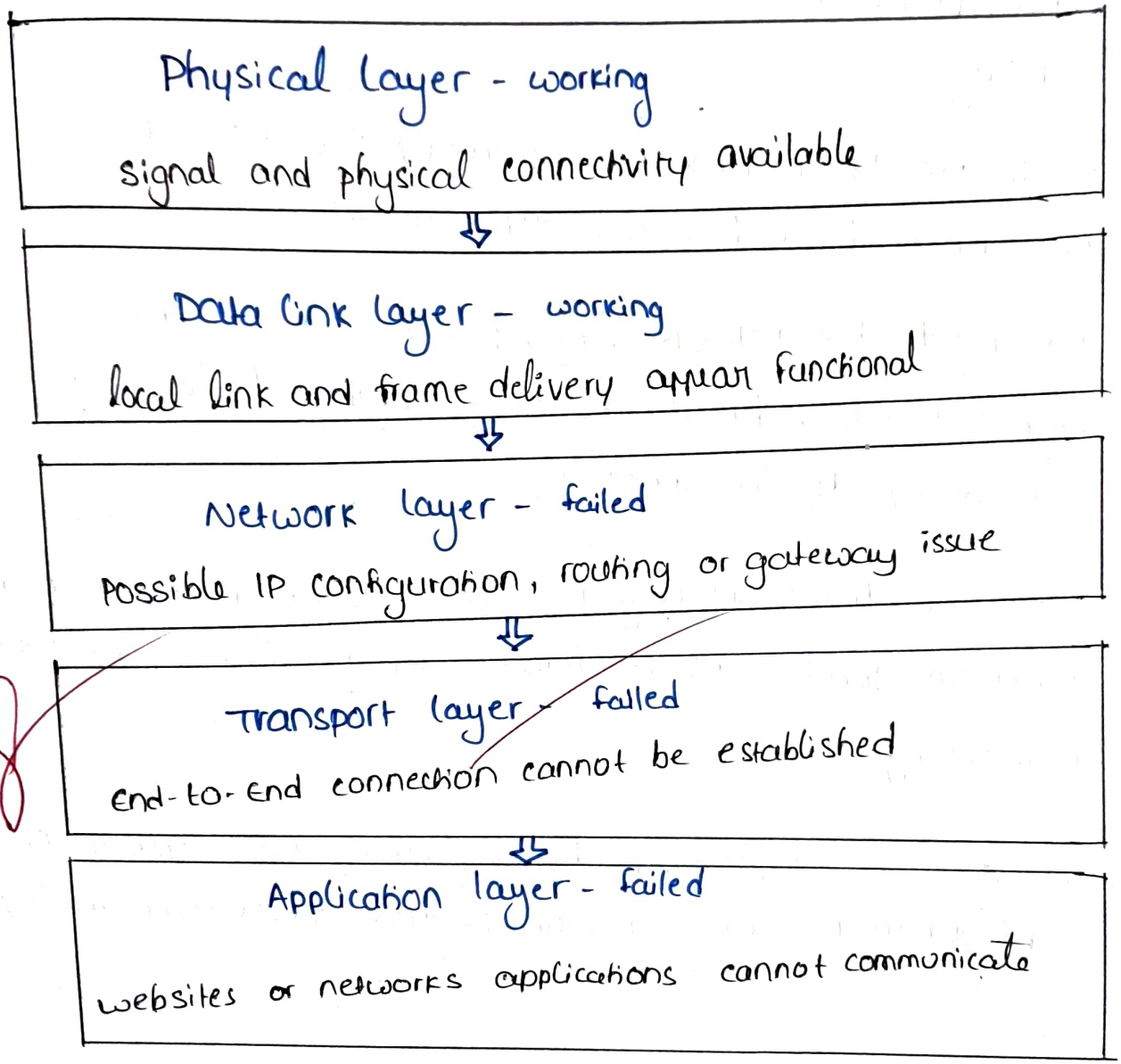
④ the Transport layer carries application data using an appropriate transport protocol.

⑤ The Network Layer routes packets toward the web server.

⑥ the Data Link layer sends frames over the local network.

⑦ the physical layer transmits the bits as signals

twork troubleshooting using the concept map.



Analysis and troubleshooting steps:

① check the physical layer: confirm the cable, Wi-Fi connection, network adapter and link indicators are working.

② check the Data link layer: verify the network interface is connected and local communication works.

③ Investigate the network layer: check the IP address, subnet mask, default gateway, routing table and connectivity to the gateway.

④ Test the transport Layer: check whether the required TCP/UDP service is reachable and whether a firewall is blocking traffic.

⑤ check the Application layer: verify the application configuration, DNS resolution and server availability.